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Deep Q-Network based Task Offloading Optimisation for Edge -IIoT System

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***Abstract*: Industrial Internet of Things (IIoT) is dynamic environment where the various sectors are involved, and enormous amounts of data are involved. The offloading of the decisions is done at the various processing units, which utilize the static models that burden the complete network. The optimisation of the parameters involved in the transmission of the offload needs to be done using various factors like latency, bandwidth, energy consumption, etc. Hence, the proposed work uses the Deep Q-network based reinforcement learning, which considers the dynamic IIoT environment factors for the evaluation of the decision based on the actions of delivering it to the local processing, edge server or cloud. The load is effectively balanced based on the offload decisions taken based on the DQN learning, and the reward value is obtained. The proposed work showed improved reward value in a positive manner with the improvement of the factors of energy consumption, latency and cost of the system, with an improvement of 94.56% of total completion rate before the deadline of the whole process.**

Keywords: IIoT, Optimisation, Innovation, DQN, edge, Offloading.

I. INTRODUCTION

Industrial Internet of Things (IIoT) uses multiple sensors of heterogeneous types to collect data from various parts of the industries and their sector for the automation process. There is the rapid growth seen in the IIoT, where huge amounts of data are involved in the process, which makes it inefficient in terms of communication because every sensor device works on the backup energy system. The communication layer is used to transfer the data from the sensor or its gateway to the cloud network, which makes it inefficient in terms of power, capacity and memory storage. Hence, the edge computing system came into the role where the IoT gateway nodes communicate with the edge computing systems nearby and then transfer it to the cloud network.

IIoT system needs to undergo the task offloading process to the high-power computing device, where the virtual support can be installed to understand the resource status, workload, and offloading decisions using the virtual twin environment that undergoes the effective optimisation technique to find the optimized value before the execution of the model [1]. The important factors involved in the offloading process that need to be optimized are the delay, energy usage, computational load and the system efficiency; hence, the local processing system can use an effective algorithm that obtains the optimized solutions that balance the factors of the IIoT devices [2].

Even though the edge resources are limited, which is a superior to the IoT nodes in terms of computing technology, the optimisation also considers the factors of edge computing to avoid the depletion of resources and to reduce the delay of communication of sensed data to the cloud network [3]. For the IIoT networks that are involved in the sensing communication-based application, it is important to consider the optimisation process in terms of a search algorithm that needs to deal with the chaos of the edge system and to undergo the reverse learning procedure that improves the convergence and the sensing reliability of the system [4].

In the IIoT system, the security-aware optimisation systems are modelled using the adversarial networks that use the data distribution to find the malicious intruders in the network that optimize the security purposes and strengthen the security by identifying the malicious model updates [5]. The main role of the IIoT systems is played by the edge computing systems and the cloud systems, where the collaborative optimisation process can be carried out between them based on reinforcement learning, which coordinates the task allocation before the offloading decisions in the IIoT environment [6].

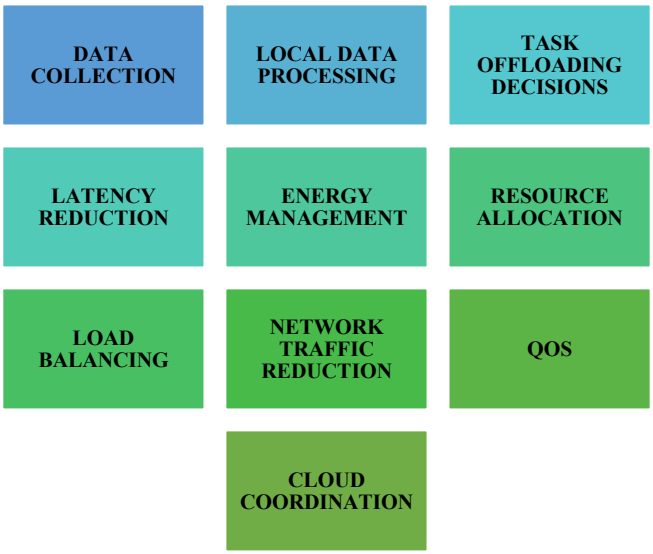


Fig. 1 Responsibility of Edge IIoT systems

In the large-scale IIoT networks, it is necessary to consider the task offload decisions before the sharing of data between the edge and the cloud because they are overloaded with data, where the collaborative learning accompanied by dynamic scheduling helps in adaptive optimisation of the system [7]. On the edge, most of the intelligent algorithms are incorporated that decide the faster rate based on the defects in the final products, where the data is obtained using the IIoT network, where the AI generation of content at the edge makes the defect detection accuracy higher [8].

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The adaptive control at the edge systems is possible by introducing the cognitive ability of the system using the machine learning technique, where the learning of patterns helps in the decision-making process based on the objective function for which the IIoT system is considered [9]. To improve the communication efficiency in the industrial IoT network, the digital twin model is incorporated that identifies the optimized solution using the virtual model that is a replica of the real-time network model, which is then used reinforcement learning to communicate it to the other side, this maximizes the throughput of the system [10].

Fig. 1 describes the important functionality of the edge IIoT systems that helps the industries for real-time monitoring and fault detection. predictive maintenance and an intelligent decision-making process. In the existing literature, the optimisation relies on the conventional factors for improvisation, where the learning-based system that adaptively optimizes the solution based on the dynamic operating conditions of the industry are not provided.

Hence, the proposed work uses the Deep Q-network based task offloading optimisation process that learns the best execution decision considering the factors that affect the performance of the Edge IIoT system are framed. Based on this proposed work, the content is framed in the manner of i) Introduction, ii) Related Content, iii) Task offloading in Edge IIoT systems, iv) Proposed Optimisation Methodology, v) Results and Discussion, and vi) Conclusion.

II. Related Content

The edge IIoT systems for the industrial applications are discussed in this section, where the different optimisation techniques that are used in the task offloading and other aspects are discussed from various literature sources.

The challenges in the edge based IIoT system includes the routing mechanism, scheduling procedures, security and the standardization that are involved in the data communication where the edge system act as the intermediate between the sensor nodes and the cloud where the intelligent systems are used for fast computation, storage, decision support and control of certain actions that needs the efficient integration of the industrial devices, edge and the cloud [11].

The dynamic conditions of the industrial activities are obtained using the contextual information that provides the relation of the sensor data obtained, which is optimized using the genetic algorithm that obtains the non-convene solution-based results following the process of selection, mutation and convergence, helping in IIoT systems in multi-objective functions [12].

The decision to process the data at the edge systems or transfer it to the cloud in the process of offloading is carried out considering the factors of task size, delay, energy and bandwidth availability of the system that uses the machine learning and reinforcement learning for making decisions using the decision parameters [13].

The distributed networks that help the IIoT system to use the microservices option during the offloading process of data that are assigned based on the resource availability, computation demand, and service requirements for the improvement of the resource usage, and it completely depends on the single computing nodes [14].

Edge systems are also limited energy-based systems where the lightweight AI models can only be used in the system that must be of less computation complexity and must have less communication overhead that needs the low power AI processing technique to produce the energy-aware Edge IIoT systems [15].

The reward-based decision actions that are done using the reinforcement learning mechanism, which helps in controlling the performance of the IIoT system that undergoes the training interactions between the model to achieve the stable learning of the dynamic environment of the industrial functions that overcomes the issue raised with the fixed or static control actions delivered in the IIoT environment [16].

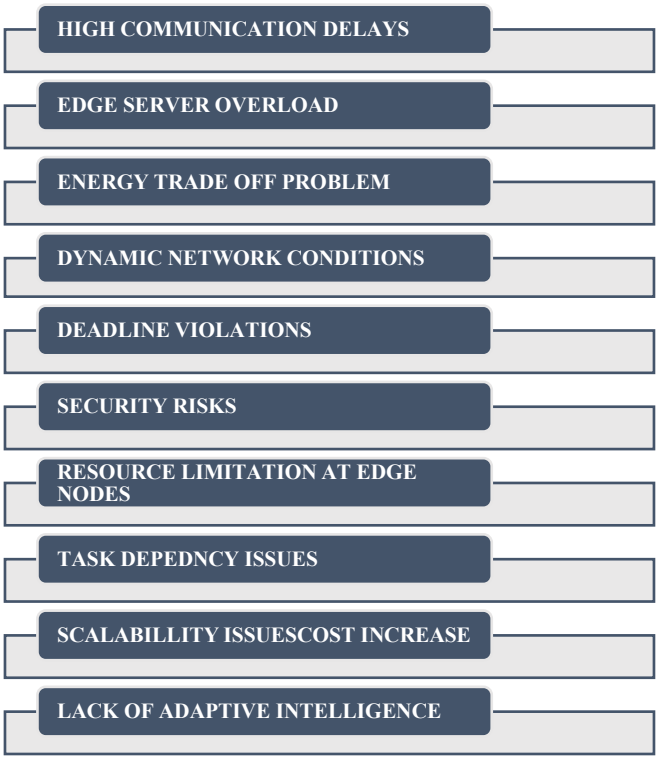


Fig. 2 Limitations of Task Offloading in Edge IIoT Systems

The efficient resource utilization between the edge and the cloud is carried out with the help of the SDN controller that supports the dynamic conditions in the industries collaboratively, like workload balancing, bandwidth allocation, and resource maintenance that uses the Pure Edge Sim [17].

The transparent management of the IIoT system is framed using the Blockchain mechanism, where the data are shared in the distributed network with a good level of authentication at the device level, which functions on the decentralized service-based management, but it has the limitation of delay-sensitive task offloading [18].

The attack detection in the IIoT system, where the deep neural network is used for reducing the dimensionality of the data, which reduces the traffic with fewer features compared to the original dimensionality. This reduces the load of computation of unwanted information; hence it is suitable as the lightweight model in the edge system due to less data size, and the security is maintained [19].

The task offloading needs an efficient 6G communication system that must consider the efficient capacity by accommodating numerous IoT devices, resulting

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in the efficient functioning of the communication layer, which reduces the bottleneck issue in the smarter resource sharing applications [20].

Fig. 2 describes the various aspects of task offloading issues in the edge IIoT system, where recent research uses a static algorithm to reduce the issues. The industrial environment is dynamic in nature; hence the system must be adaptive enough for task offloading operations.

III. Task Offloading in Edge IIoT System

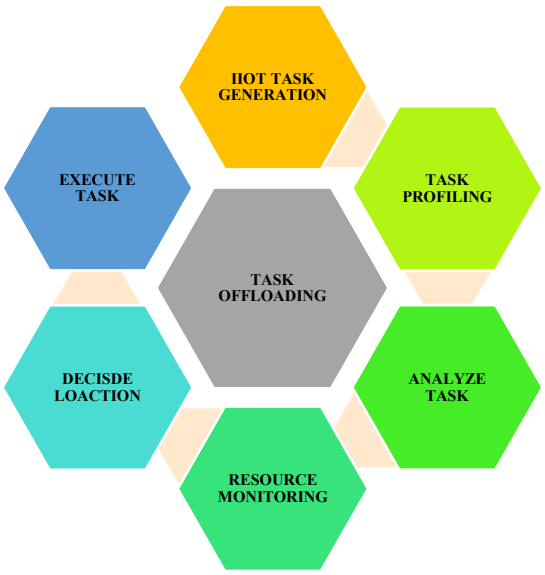


Fig. 3 Process of Task Offloading

Task offloading is the process that is undergone at the different levels of the IIoT system, at the sensor level, edge level and in the cloud level. The task needs to be processed in the local system that obtains the decision based on the availability of the system. Fig. 3 discusses the various processes involved in the task offloading at the edge level of the IIoT system [21].

IIoT system obtains the data from the sensor devices, robots, PLX cameras and the generated tasks, such as vibration monitoring, fault detection, quality inspection, etc. The analysis of the task needs to be accessed as shown in Fig. 4, which considers the task size, CPU cycles, deadline, security, task priority, energy demand of the systems, etc., based on which the task profiling is carried out.

Offloading decisions are carried out based on the device CPU capacity, server load, cloud availability, bandwidth, queue length, network delay that provides the insights into the available resources for the Edge IIoT system. The offloading decision-making process is carried out with the help of the neural network model that predicts the size of the task and assigns the execution level at local, edge and cloud.

In the task execution level, the local devices are used to process the simple and urgent tasks, the edge server is used to process the delay-sensitive industrial tasks and the cloud server is used to process the heavy tasks based on the strict deadlines. The fault detection, status of the machine, control signal and the maintenance alert are obtained after the processing, which is sent back to the IIoT device or the industrial controller [22].

Performance metric of the offloading is calculated, that are based on the latency, energy consumption, task completion rate, cost deadline risks, utilization of the edge server, and the reward value the system evaluates, which helps in knowing the status of the system. This evaluation of the metrics helps in evaluating the decisions and offloading actions, which helps in improving the system's decision-making [23].

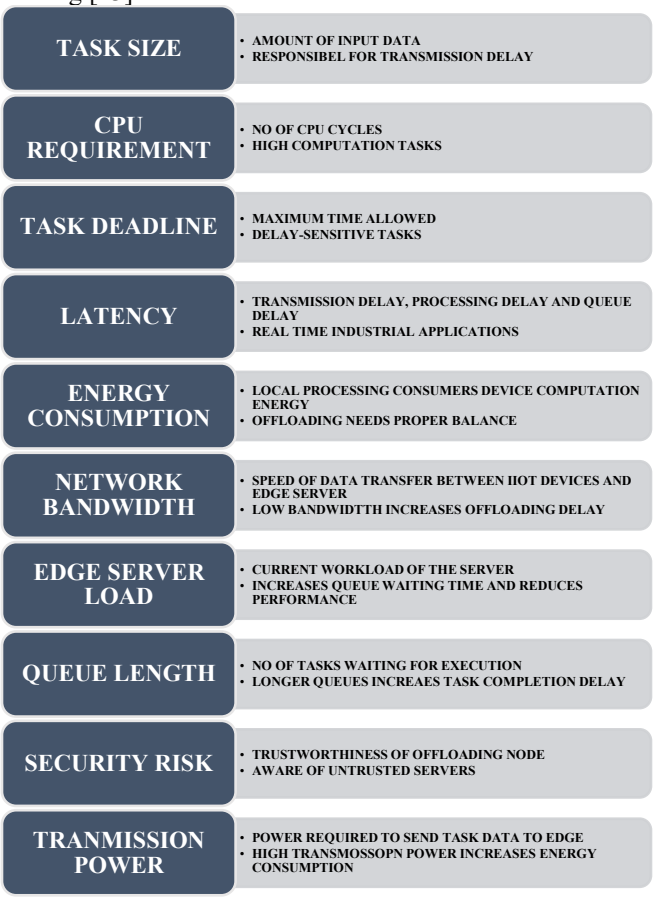


Fig. 4 Factors of Task Offloading

IV. Proposed Optimisation Methodology

The proposed work considers the dynamic conditions of the industrial environment where the data load varies based on the situation the sensors are activated to obtain the data. For the fault detection systems, if an emergency arises, then it is important to obtain the data from the sensors that are unusual pattern, and a large amount of data will be collected to verify the emergency consideration. This makes the dynamic environment for the industrial actions, hence the proposed work shown in Fig. 5 uses the Deep Q network as the reinforcement learning methodology to obtain the rewards based on which the offloading operations will be carried out.

Input to the DQN agents is the decisions made by the neural network that decides the conditions like fault occurrence, quality inspection results and the predictive maintenance activities. These activity-based decisions are obtained using the Long Short-Term Memory (LSTM) network, which works based on the time series data from the sensor network of the machines in the industrial applications.

The important aspect of the task offloading is the task generation, where the proposed work considers the tasks from the decision made using the LSTM actions, such as task

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size, CPU cycles, deadline, and priority. The generated tasks are then classified as the delay-sensitive task, computation-intensive task, energy- sensitive task, priority based takes and the normal monitoring task. It is important to collect the task profiling of the generated tasks, which is an important process where the importance of the processing location of the offloading operations is determined. Usual task features that are considered in the task profiling are the input data size, CPU cycles, task deadlines, task priority, output data size, and execution complexity.

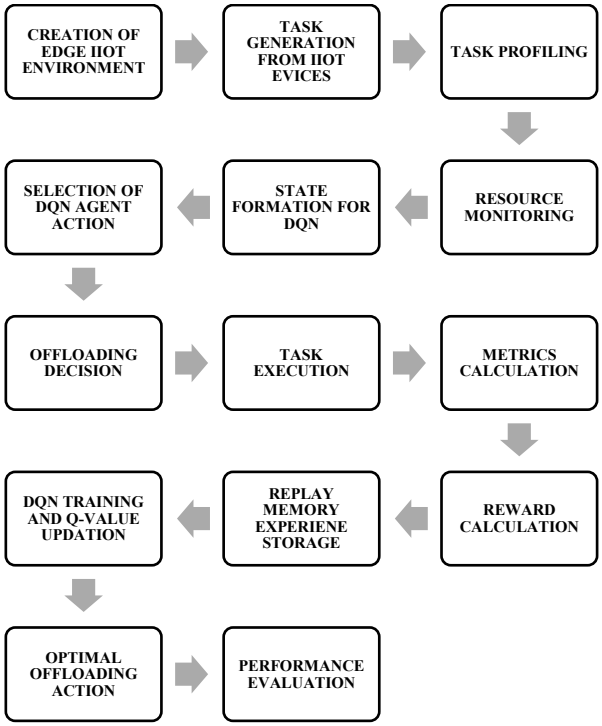


Fig. 5 Proposed Task Offloading Framework

The resources in the edge IIoT networks are monitored on the basis of evaluation of the available bandwidth, edge server load, queue length, device energy level, etc., which are carried out continuously to understand the current situation of the resources. The DQN operation is shown in Fig. 6, where the state formation is obtained, which is the collection of the task and resource formation that are represented as vectors, which helps to learn the current task-based demand and the resources that are available in the edge systems.

In the DQN agent selection, the possible actions chosen are 0, 1, and 2, that exhibits the results of the local execution, edge server execution, and the cloud server execution. This action is selected based on the different edge -IIoT conditions that trigger the different offloading actions, and based on that, it selects the learned actions. If the load is less than, the local processing is used, edge is used for moderate offloads, and the cloud is chosen for the heavy data load.

Based on the decision, the task transmission is carried out, where the transmission, which is the ratio of the task data size and the bandwidth, is calculated along with the transmission energy that is obtained by the transmission power per transmission time period, which helps to know both the delay and energy consumption. The period of the task execution is calculated by the ratio of the CPU cycles by the CPU frequency, where the calculation differs for the local execution, edge execution and the cloud execution [24]. Once the execution is carried out, the results are sent to the IIoT devices, where the results must reach the destination before the task deadline, based on which the performance metrics of the system are obtained [25]. Using this, the total task completion delay of the proposed system is obtained using (1),

$$T_i = x_i^L T_i^L + x_i^E T_i^E + x_i^c T_i^c \quad (1)$$

In (1'0, the total completion of the delay of task 'i' is obtained using the decision variables of offloading x_i^L, x_i^E, and x_i^C based on its task completion time T_i^L, T_i^E, and T_i^C.

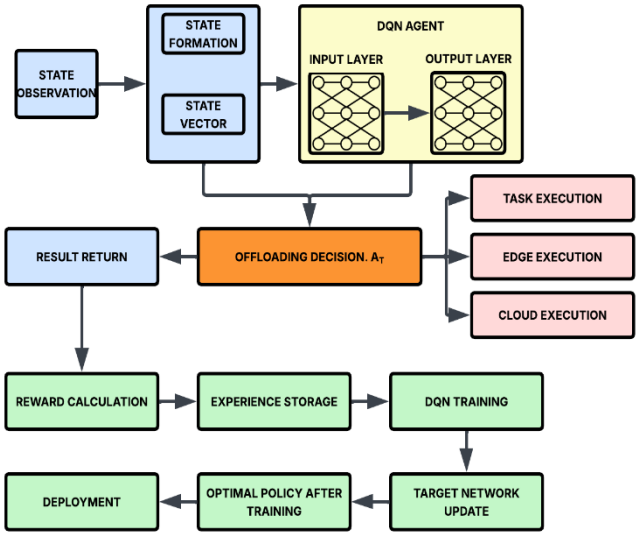


Fig. 6 DQN Operation

Based on the IIoT devices' reward function calculation based on the offloading decision that is offered by the DQN network, the results are varied, where the formulation is given in (2).

$$R_i = -(\alpha T_i + \beta E_i + \gamma C_i + \delta V_i) \quad (2)$$

The reward for the task 'i' is R_i that depends on the task completion delay T, energy consumption E_i, computation cost C_i, and the deadline violation penalty V_i that uses the weight parameters α, β, and γ. The DQN stores each experience in the replay memory, which allows the DQN to learn from the older offloading decisions for the next state. Now the proposed system undergoes the DQN training process that includes the selection of a mini-batch, calculating the predicted Q-value and target Q-value, and obtaining the final loss value that updates the neural network weights. The Q-value updates are given in (3),

$$Target_Q = R + \gamma \max (S_{next} + A_{next}) \quad (3)$$

The loss function is calculated using the formulation (4).

$$loss = (Q_{Target} - Q_{Predicted})^2 \quad (4)$$

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Using (4), the DQN agent learns the best offloading policy that is updated in the target network for the stabilization of the DQN training.

V. Results and Discussion

The task offloading decision based on the payload and the dynamic conditions of the IIoT environment is carried out using the DQN learning process that obtains the rewards for updating its policy, which makes the edge IIoT system efficient.

The parameters used are listed in Table I, and the Edge IIoT dataset from Kaggle, which contains the network traffic features, is selected, providing the features of device communication, traffic behaviour, and the environmental conditions.

TABLE I. Simulation Parameters

PARAMETERS	VALUES
TASK SIZE	100 KB
CPU CYCLES	500MEGACYCLE
LOCAL CPU FREQ	1.5GHZ
EDGE CPU FREQ	50 GHZ
CLOUD CPU FREQUENCY	100GHZ
BANDWIDTH	50MBPS
TRANSMISSION POWER	2W
REPLAY MEMORY SIZE	10000
LEARNING RATE	0.0001
DISCOUNT FACTOR	0.95
BATCH SIZE	32

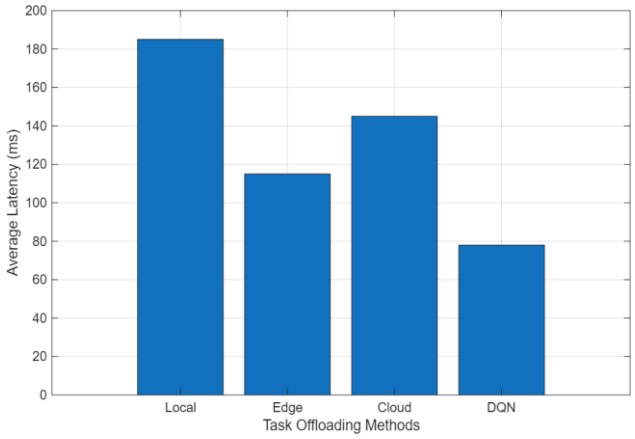


Fig. 7 Average Latency for Task Offloading Methods

Fig. 7 is the average latency of the proposed work based on the task offloading methods white the local processing is high where the proposed DQN based has the low DQN value of 72ms where the conditions is taken under dynamic conditions.

Fig. 8 is the energy consumption, where the local processing in the cloud has higher energy consumption, but the DQN has the lowest energy consumption with 9J, which is nearly 14J less than the highest energy consumed.

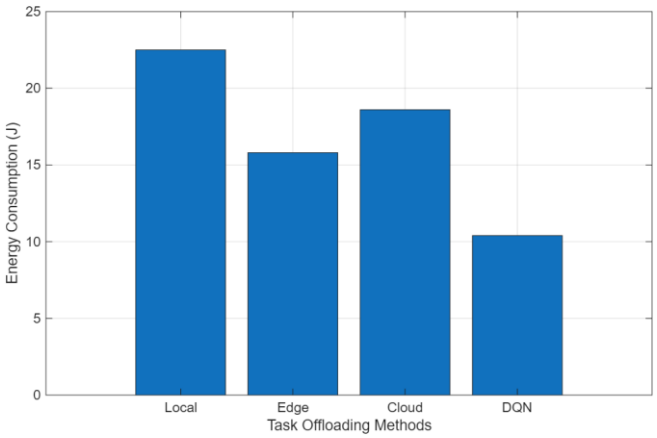


Fig. 8 Energy Consumption for Task Offloading Methods

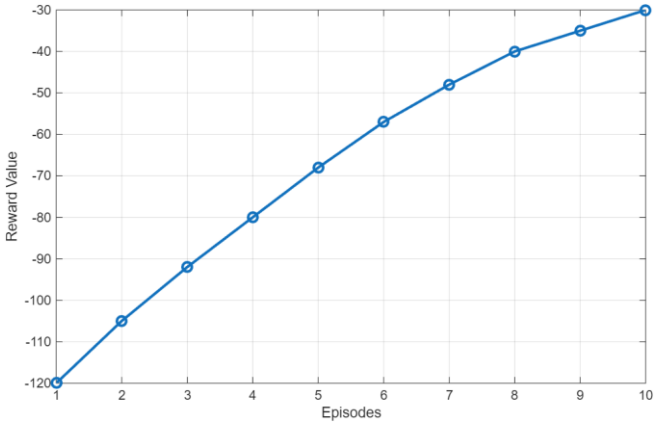


Fig. 9 Reward Value for the Proposed Work

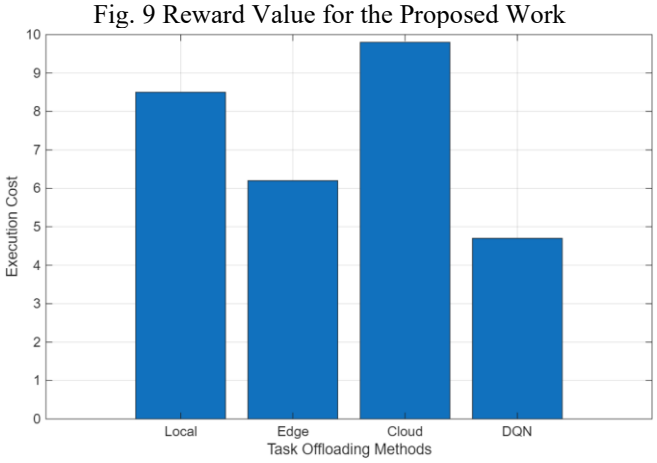


Fig. 10 Execution Cost for Task Offloading Methods

Fig. 9 is the reward value obtained for the DQN actions taken. It is seen that the reward value grows from -120 to -30, where the value rises. This shows that the utilized DQN process effectively obtained it. This shows the dynamic assignment of task offload at the edge, local, and cloud, enhancing the system performance.

Fig. 10 shows the performance of the execution cost, where the cloud server shows the highest cost, while the DQN shows less, which is less than that of the edge computing systems.

Fig. 11 shows the plot of the task completion rates of the different edge, local and cloud, which is compared with the proposed DQN model, where the performance increased

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to 94.56%, which is 9% higher than that of the efficient edge computing systems.

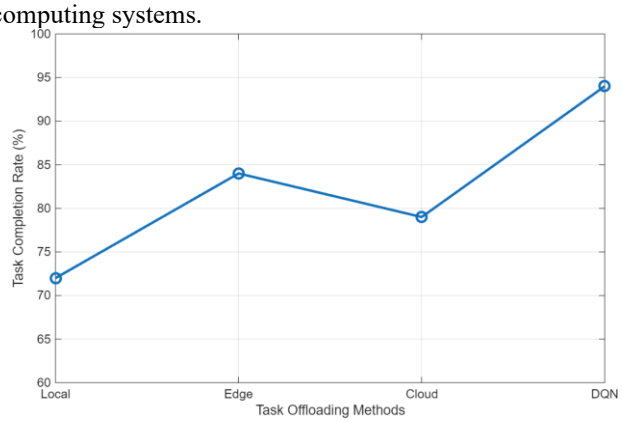


Fig. 11 Average Latency for Task Offloading Methods

VI. Conclusion

The proposed work undergoes the task offloading operation for the Edge-based IIoT systems, where the reinforcement-based DQN learning method is used. The dynamic conditions of the IIoT environment are learned by forming the resource allocation and the state formation. Using the three different actions at local processing, edge and the cloud, the offloading is carried out based on DQN decisions. The proposed work evaluated the reward function that grows with the performance increment of 94.56% based on the total completion rate of the task offloading at the local processing, edge and cloud.

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